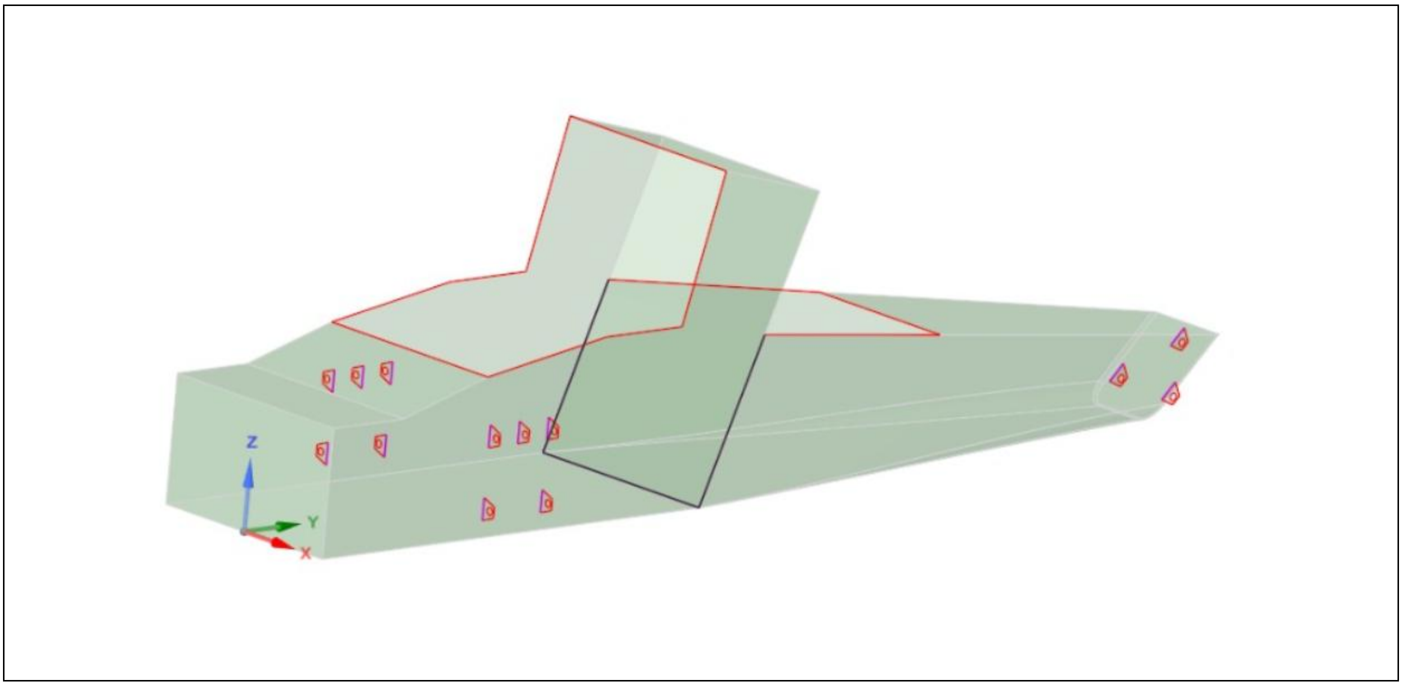
PFEM MARATHON 2025 GROUP 6 REPORT

Monocoque Tail Design



Members

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Desing Modifications

The following design modifications were made compared to the demo model.

- The tail rear wall was modified with a slope of 45 deg (Fig1) for better alignment of rear suspension.
- The rear suspension was offset in the demo model. To reduce torsional deformations the suspension was shifted to be symmetric(Fig2).
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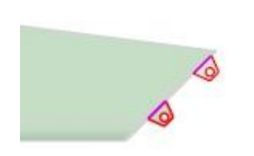

Fig:1

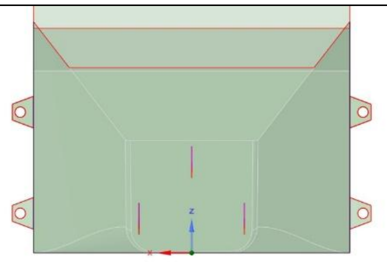

Fig: 2

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Final Parameter Selection

Using this model several iterations were run in Structural analysis with the given parameters as in the demo.

Final parameters chosen based on the iterations are given below:

Chassis Length	3220 mm
Tail Bottom Width	245 mm
Tail Upper Width	250 mm
Tail height	213 mm
Tail Cap Length	1083.7 mm
Chassis Mass	16.126 kg

For determination of the above parameters direct optimization module was utilized optimized parametric ranges. A screenshot of the results is shown below (Fig 3 and 4).

Table of Schematic D2: Optimization

	A	B	C	D
1	Name	P19 - cap_height	P21 - Output Parameter (N m rad^-1)	P22 - Joint Probe Y Axis (degree)
2	1	1093.8	30091	0.0019041
3	2	1101.3	30428	0.001883
4	3	1108.8	30770	0.0018621
5	4	1116.3	31111	0.0018417
6	5	1123.8	31460	0.0018212
7	6	1131.3	31808	0.0018013
8	7 DP 80	1138.8	32157	0.0017817
9	8	1146.3	32511	0.0017624
10	9	1115.4	31067	0.0018442
11	10	1115.4	31068	0.0018442
12	11	1116.2	31106	0.001842

Fig:3

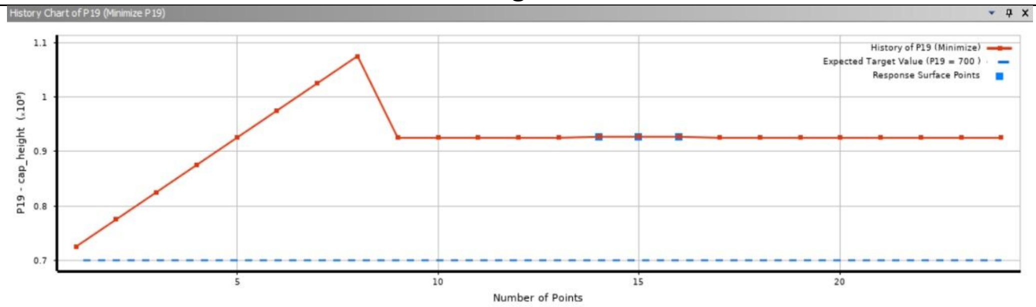

Fig:4

Analysis and Validation

The following are the results from Analysis:

Under Normal Loading conditions of point mass 230kg and under std gravity.

Torsional Stiffness	30162 Nm/rad
Spring Probe Elongation	-9.01982576 mm
Bending def Z Axis	0.468869288 mm
Rear Chassis Mass	16.12595436 kg

Under 3g load during turning and acceleration the following values of ground clearance was reported. It can be seen that the clearance criteria of -90 mm are met.

Ground Clearance in turning	-18.084 mm
Ground Clearance in acceleration	-6.465 mm

Harmonic Analysis

As majority of the road expiation frequency is in the range of 10-20Hz harmonic analysis in this range were carried out under 1g load in all the bodies. The maximum deflection due to resonance was within the range f ground clearance.

Natural frequency of the body	5.089 Hz
Max deflection in the z axis	-10.947 mm

Non-Parametric Approach

A non-parametric design approach was explored in which to improve the torsional stiffness with optimized mass the parameters where fixed with the following values:

Tail length	2000 mm
Tail Upper Width	500 mm
Tail Lower Width	400 mm
Tail height	300 mm
Spring Stiffness	90 N/mm

In the previous parametric cases it was seen that the torsional stiffness values determined was close to the limit of 520 Nm/deg or 30000Nm/rad. So, to resolve this issue Topology optimization was carried out with just the Chasis under torsional moment 1Nm on the front edge while fixing the rear edge to determine the topology which can be optimized for lowest mass. This gave us a general idea where the stress was lowest and how topology can be trimmed to reduce mass. From the structural optimized figure, another tail section was designed taking hints from it.

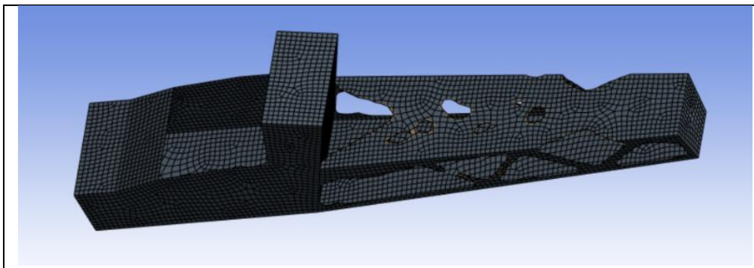

Fig 5: Structural Optimization

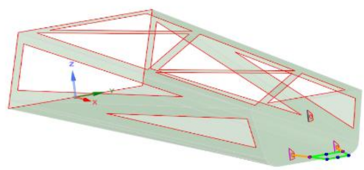

Fig 6: Trimmed Tail section

Then from the New Structure Static Structural simulation was run again. Now the torsional Stiffness was

1465 Nm/deg with the mass of the tail segment = 15.934 kg. This structure was simulated under maximum 3g loads in braking, acceleration and turning. The following were the outcomes:

under 3g braking		under 3g acc		under 3g turning	
ground clearance	50 mm	ground clearance	85 mm	ground clearance	84 mm
Max Stress	113.25 Mpa	Max Stress	46.19Mpa	Max Stress	75 Mpa
bending probe	0.1361 mm	bending probe	0.013 mm	bending probe	0.56 mm
spring deff.	4.07 mm	spring def.	0.54 mm	spring def.	3.8 mm

From the results it can be seen the following requirements were met:

- Ground Clearance of 90mm maintained
- Chassis bending deflection is below 20% of the spring deflection
- Torsional Deflection is above 520 Nm/deg

Here are few snippets from our analysis –

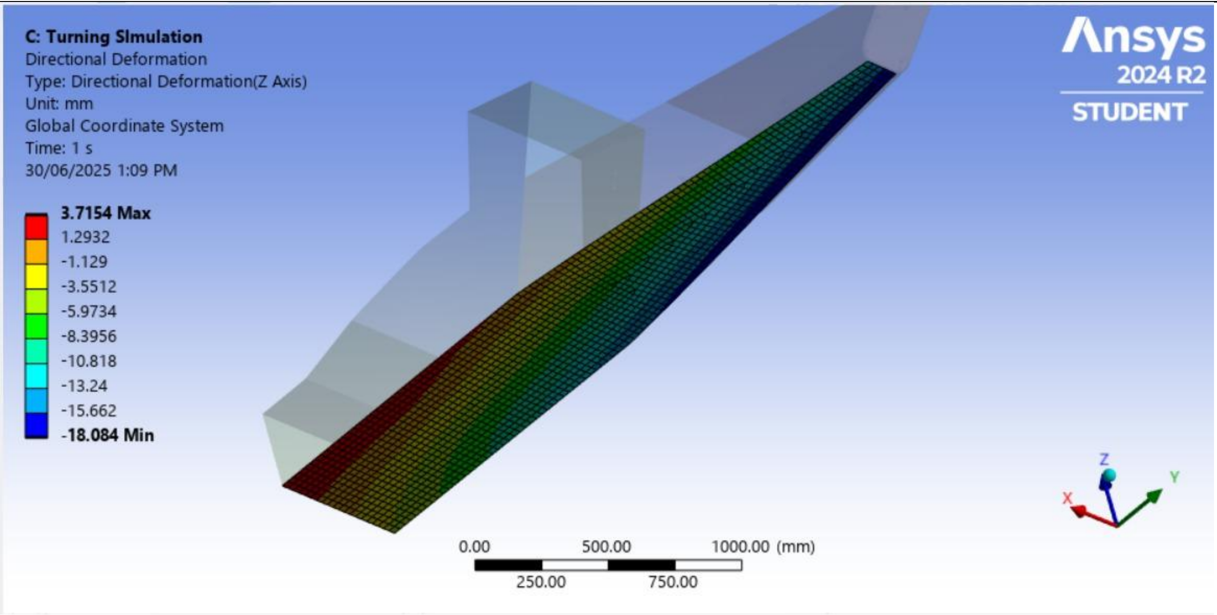

Fig 7: Ground clearance During Turning

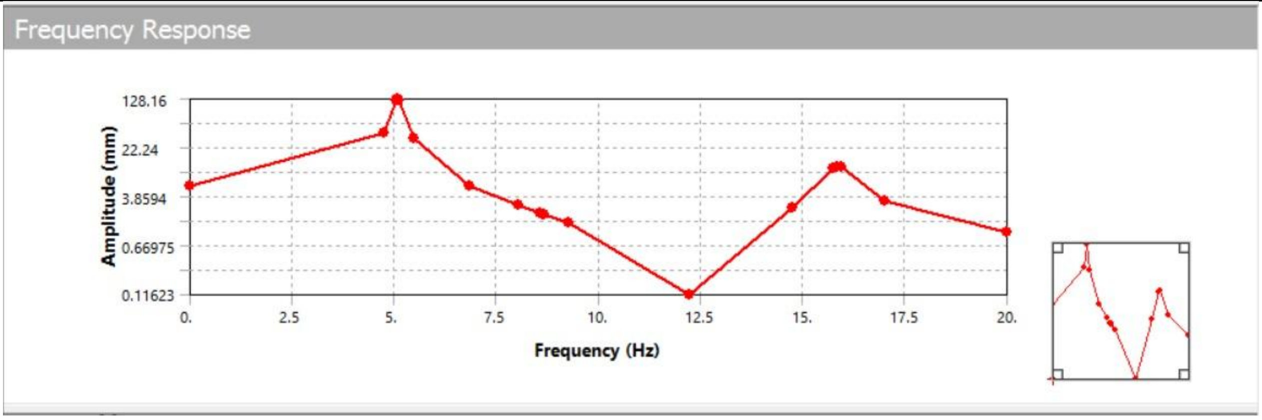

Fig 8: Harmonic Analysis Response

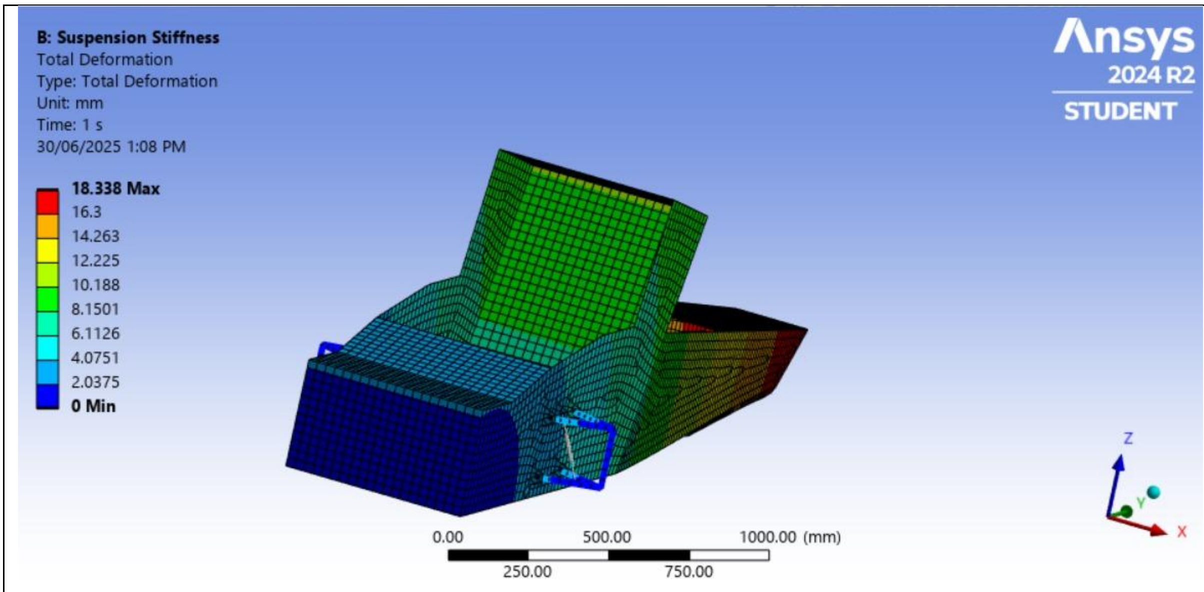

Fig 9: bending Stiffness

Here are few snippets of our finalized chassis design post validation.

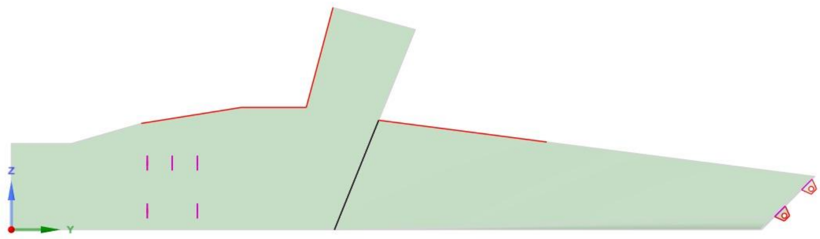

Figure 10: Final Design Side view

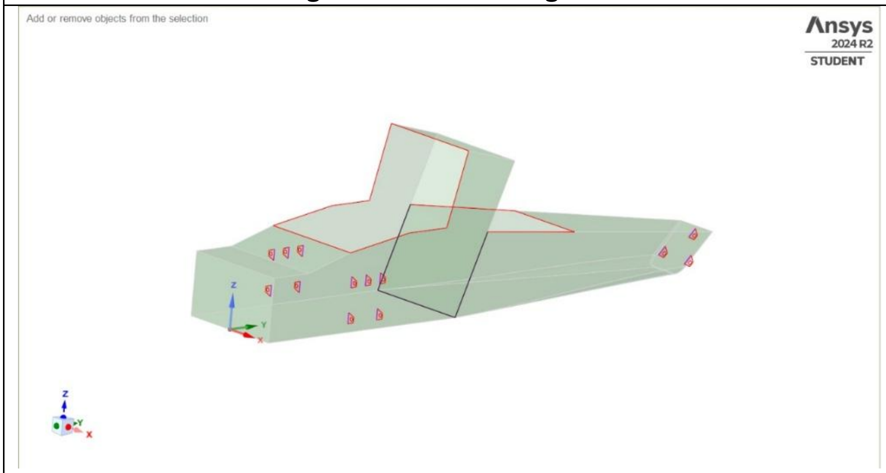

Fig 11: iso view

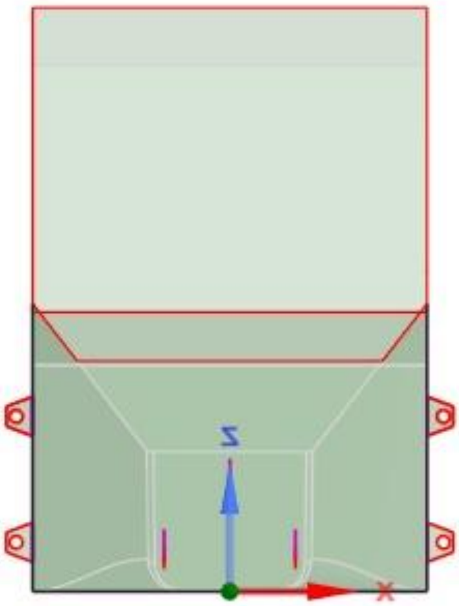

Fig 12: rear View